

CLASS-8

LESSON-12 FACTORISATION

(This PDF Based on NCERT Book)

FACTORS OF NATURAL NUMBER(प्राकृतिक संख्याओं के गुणनखंड)-

A **factor** is a number that divides another number completely, leaving **no remainder**.

How it Works

When you multiply two natural numbers together, the result is called a **product**. The two numbers you multiplied are the **factors** of that product.

Number $\times$ Number = Product

(Factor) $\times$ (Factor) = Product

Example: Factors of 12

To find the factors of 12, look for all the pairs of numbers that multiply to make 12: $\div \times$

- $1 \times 12 = 12$
- $2 \times 6 = 12$
- $3 \times 4 = 12$

So, the factors of 12 are: **1, 2, 3, 4, 6, and 12.**

Key Rules to Remember

- **1 is a factor of every number:** Every natural number can be divided by 1.
- **The number itself is always a factor:** For example, 50 is a factor of 50 because $50 \div 50 = 1$.
- **Factors are never larger than the number:** A factor of 10 cannot be 11; it must be 10 or smaller.
- **Finite count:** Unlike multiples (which go on forever), a number has a limited, specific list of factors.

FACTOR OF ALGEBRAIC EXPRESSIONS(बीजीय व्यंजकों का गुणनखंड)-

Finding the factors of an **algebraic expression** is very similar to finding factors of a regular number. It simply means breaking an expression down into a multiplication problem.

In simple words, **factoring** is the opposite of multiplying.

1. The Core Idea

If you multiply $2(x + 3)$, you get $2x + 6$.

Factoring is just starting with $2x + 6$ and figuring out that it came from $2 \times (x + 3)$.

- **Expression:** $2x + 6$
- **Factors:** 2 and $(x + 3)$

2. How to Factor (The "Common Factor" Method)

The easiest way to factor is to look for something that "fits" into every part of the expression.

Example: Factor $5x + 10$

1. **Look at the numbers:** What is the largest number that divides into both 5 and 10?
 - The answer is 5.
2. **Pull it out:** Put that number outside of a set of parentheses.
 - 5
3. **Fill in the blanks:** Divide each part of the original expression by 5.
 - $5x \div 5 = x$
 - $10 \div 5 = 2$
4. **The Result:** $5(x + 2)$

3. Factoring Variables

You can also pull out letters (variables) if they appear in every term.

Example: Factor $x^2 + 3x$

- Both terms have at least one x.
- Pull one x out: $x \dots$
- Divide x^2 by x, you get x.
- Divide $3x$ by x, you get 3.
- **The Result:** $x(x + 3)$

Why do we do this?

Factoring is like "untangling" a knot. It makes complicated equations much easier to solve, especially when you are trying to find the value of x or simplify a large fraction.

Term	Simple Meaning
Expansion	Multiplying to remove brackets.
Factoring	Dividing to create brackets.

WHAT IS FACTORISATION(गुणनखंड क्या है)-

Factorization is the process of breaking a number or an expression down into smaller parts that, when **multiplied together**, give you the original result.

The "Reverse" Rule

Factorization is just **multiplication in reverse**.

- **Multiplication:** You start with small pieces and combine them.
 - $3 \times 5 = 15$
- **Factorization:** You start with the total and find the pieces.
 - $15 = 3 \times 5$

In algebra, it works the same way:

- **Multiplication:** $x(x + 2) = x^2 + 2x$
- **Factorization:** $x^2 + 2x = x(x + 2)$

METHOD OF COMMON FACTORS(सामान्य गुणखंड की विधि)-

The **Method of Common Factors** is the simplest way to factorize an expression. It involves finding a number or a variable (letter) that is present in **every term** of the expression and "pulling it out."

Step-by-Step Process

To use this method, follow these three steps:

1. Find the Highest Common Factor (HCF)

Look at all the terms and find the largest number and the highest power of each variable that divides into all of them.

2. Divide each term

Divide every part of your original expression by that HCF to see what is left over.

3. Write the result

Write the HCF outside a set of brackets, and put the "leftovers" inside the brackets.

Example 1: Numbers Only

Factorize: $12x + 18$

- **Step 1:** What is the biggest number that goes into both 12 and 18? It is **6**.
- **Step 2:** Divide both by 6.
 - $12x \div 6 = 2x$
 - $18 \div 6 = 3$
- **Step 3:** Put it together: **$6(2x + 3)$**

Example 2: Variables (Letters)

Factorize: $x^2 + 5x$

- **Step 1:** What do both terms have? They both have at least one **x**.
- **Step 2:** Divide both by **x**.
 - $x^2 \div x = x$
 - $5x \div x = 5$
- **Step 3:** Put it together: **$x(x + 5)$**

Example 3: Numbers and Variables

Factorize: $4a^2 - 8ab$

1. **Numbers:** The HCF of 4 and 8 is **4**.
2. **Letters:** Both terms have an **a**.
3. **Total HCF:** **4a**.
4. **Divide:**
 - $4a^2 \div 4a = a$
 - $-8ab \div 4a = -2b$
5. **Final Answer:** **$4a(a - 2b)$**

FACTORISATION BY REGROUPING TERM(पुनर्समूहीकरण द्वारा गुणनखंडन)-

Step-by-Step Process

To use this method, follow these three steps:

1. Find the Highest Common Factor (HCF)

Look at all the terms and find the largest number and the highest power of each variable that divides into all of them.

2. Divide each term

Divide every part of your original expression by that HCF to see what is left over.

3. Write the result

Write the HCF outside a set of brackets, and put the "leftovers" inside the brackets.

Example 1: Numbers Only

Factorize: $12x + 18$

- **Step 1:** What is the biggest number that goes into both 12 and 18? It is **6**.
- **Step 2:** Divide both by 6.
 - $12x \div 6 = 2x$
 - $18 \div 6 = 3$
- **Step 3:** Put it together: **$6(2x + 3)$**

Example 2: Variables (Letters)

Factorize: $x^2 + 5x$

- **Step 1:** What do both terms have? They both have at least one **x**.
- **Step 2:** Divide both by **x**.
 - $x^2 \div x = x$
 - $5x \div x = 5$
- **Step 3:** Put it together: **$x(x + 5)$**

Example 3: Numbers and Variables

Factorize: $4a^2 - 8ab$

1. **Numbers:** The HCF of 4 and 8 is **4**.
2. **Letters:** Both terms have an **a**.
3. **Total HCF:** **4a**.
4. **Divide:**
 - $4a^2 \div 4a = a$
 - $-8ab \div 4a = -2b$
5. **Final Answer:** **$4a(a - 2b)$**

FACTORISATION USING IDENTITIES(सर्वसमिकाओं का उपयोग करके गुणनखंड)-

Using identities is like using a **shortcut** or a **template**. Instead of doing a lot of division, you recognize a specific pattern and immediately know what the factors are.

There are three main identities (patterns) we use for factorization.

1. The Square of a Sum

Pattern: $a^2 + 2ab + b^2 = (a + b)^2$

Use this when you have three terms, and the first and last terms are perfect squares, with a positive middle term.

Example: Factorize $x^2 + 6x + 9$

- The first term is x^2 (which is $(x)^2$)
- The last term is 9 (which is $(3)^2$)
- The middle term is $2 \times x \times 3 = 6x$
- **Answer:** $(x + 3)^2$ or $(x + 3)(x + 3)$

2. The Square of a Difference

Pattern: $a^2 - 2ab + b^2 = (a - b)^2$

This is exactly like the first one, but the middle term has a **minus** sign.

Example: Factorize $4x^2 - 20x + 25$

- First term: $(2x)^2$
- Last term: $(5)^2$
- Middle term: $2 \times (2x) \times (5) = 20x$
- Since the middle is minus, we use $(a - b)^2$
- **Answer:** $(2x - 5)^2$

3. Difference of Two Squares (Very Common!)

Pattern: $a^2 - b^2 = (a + b)(a - b)$

Use this when you have only **two terms**, both are perfect squares, and there is a **minus** sign between them.

Example: Factorize $x^2 - 16$

- First term: $(x)^2$
- Last term: $(4)^2$
- **Answer:** $(x + 4)(x - 4)$

Example: Factorize $49p^2 - 81$

- $(7p)^2 - (9)^2$
- **Answer:** $(7p + 9)(7p - 9)$

How to choose the right identity?

If the expression looks like...	Use this Identity
3 terms, all plus (+)	$(a + b)^2$
3 terms, middle is minus (-)	$(a - b)^2$
2 terms, separated by minus (-)	$(a + b)(a - b)$

FACTORS OF THE FORM $(x+a)(x+b)$ के रूप में गुणनखंड:-

This is a special case of factorization where we look at a quadratic expression (usually starting with x^2) and try to break it back down into two brackets.

This method is based on the identity:

$$(x + a)(x + b) = x^2 + (a + b)x + ab$$

To factorize an expression like $x^2 + 5x + 6$ into the form $(x + a)(x + b)$, you have to play a little "number game."

The Number Game Rules

To find the values of **a** and **b**, you need two numbers that satisfy two conditions at the same time:

1. **Multiply** to give the last number (the constant term, ab).
2. **Add** to give the middle number (the coefficient of x , which is $a + b$).

Example 1: $x^2 + 5x + 6$

- **Identify the targets:** We need two numbers that multiply to **6** and add up to **5**.
- **List pairs that multiply to 6:**
 - $1 \times 6 = 6$ (But $1 + 6 = 7$ — *No*)
 - $2 \times 3 = 6$ (And $2 + 3 = 5$ — **Yes!**)
- **The factors are:** $a = 2$ and $b = 3$.
- **Final Answer:** $(x + 2)(x + 3)$

Example 2: Dealing with Minus Signs ($x^2 - 7x + 10$)

- **Identify the targets:** Multiply to **+10**, add to **-7**.
- **Think:** Since they multiply to a positive but add to a negative, both numbers must be negative.
- **List pairs:**
 - $-1 \times -10 = 10$ (Add to -11 — *No*)
 - $-2 \times -5 = 10$ (Add to -7 — **Yes!**)
- **Final Answer:** $(x - 2)(x - 5)$

Example 3: Different Signs ($x^2 + 2x - 8$)

- **Identify the targets:** Multiply to **-8**, add to **+2**.
- **Think:** To multiply to a negative, one number must be (+) and one must be (-).
- **List pairs:**
 - $+1, -8$ (Add to -7)
 - $-1, +8$ (Add to $+7$)
 - $-2, +4$ (Add to $+2$ — **Yes!**)
- **Final Answer:** $(x - 2)(x + 4)$

Summary Checklist

When you see $x^2 + px + q$:

1. Look at the **last number (q)**. Find all pairs that multiply to it.
2. Look at the **middle number (p)**. Find which pair adds up to it.
3. Write them in the brackets: $(x + \text{first number})(x + \text{second number})$.

DIVISION OF ALGEBRAIC EXPRESSIONS(बीजीय व्यंजकों का विभाजन)-

Division of algebraic expressions is essentially the opposite of multiplication. When we divide, we are trying to find out how many times one expression "fits" into another, or we are simply reducing a fraction.

There are three main ways to handle division, depending on what you are dividing by.

1. Dividing a Monomial by a Monomial(एकपदी का दूसरे एकपदी से विभाजन)

A "monomial" is just a single term (like $6x^2$). To divide, you handle the numbers and the variables separately.

Rule: Divide the coefficients (numbers) and subtract the exponents of the same variables.

$$\frac{x^a}{x^b} = x^{a-b}$$

Example: Divide $15x^4$ by $3x^2$

- **Numbers:** $15 \div 3 = 5$
- **Variables:** $x^4 \div x^2 = x^{(4-2)} = x^2$
- **Result:** $5x^2$

2. Dividing a Polynomial by a Monomial(एकपदी द्वारा बहुपदी का विभाजन)

When you divide a long expression (polynomial) by a single term, you must divide **every single term** of the polynomial by that term.

Example: Divide $(8x^3 + 4x^2 - 2x)$ by $2x$

$$1. \frac{8x^3}{2x} = 4x^2$$

$$2. \frac{4x^2}{2x} = 2x$$

$$3. \frac{-2x}{2x} = -1$$

- **Result:** $4x^2 + 2x - 1$

3. Dividing a Polynomial by a Polynomial (Using Factorization)(एक बहुपद का दूसरे बहुपद से भाग देना)

The easiest way to divide a long expression by another expression (like $x+2$) is to **factorize first**. Often, a common factor will appear in both the top and the bottom, allowing you to "cancel" them out.

Example: Divide $(x^2 + 5x + 6)$ by $(x + 3)$

1. **Factorize the top:** We know from our previous lesson that $x^2 + 5x + 6 = (x + 2)(x + 3)$.
2. **Write as a fraction:**

$$\frac{(x+2)(x+3)}{(x+3)}$$

3. **Cancel the common term:** Since $(x + 3)$ is on both the top and the bottom, they cancel each other out.
4. **Result:** $x + 2$

Key Things to Remember

- **The "Zero" Rule:** You can never divide by an expression that equals zero.
- **Signs Matter:** Be careful with negative signs. Dividing a negative by a negative makes a positive, just like in regular division.
- **Simplification:** Think of division as "simplifying a fraction." If you can't divide perfectly, you might just leave it as a simplified fraction.